Python Lists

Creating a List

```
cart = ["T-shirt", "Lamp", "Pen"]
print(cart)

# A list of mixed data types
my_list = [1, "Python", 3.14]
print(my_list)

# Empty list
my_list = []
print(my_list)
```

```
['T-shirt', 'Lamp', 'Pen']
[1, 'Python', 3.14]
[]
```

Convert a string to a list

```
vowels = "aeiou"
vowels_list = list(vowels)
print(vowels_list)
```

```
['a', 'e', 'i', 'o', 'u']
```

Accessing List Items

```
languages = ["Python", "Swift", "C++"]

# Access the first item
print(f"languages[0] = {languages[0]}")

# Access the third item
print(f"languages[2] = {languages[2]}")
```

```
languages[0] = Python
languages[2] = C++
```

Negative Indexing

```
languages = ["Python", "Swift", "C++"]

# Access the last item
print('languages[-1] =', languages[-1])

# Access the third last item
print('languages[-3] =', languages[-3])
```

```
languages[-1] = C++
languages[-3] = Python
```

Adding and Updating Items

Updating List Items

```
cart = ["T-shirt", "Lamp", "Pen"]

# Update second item to "Shoes"
cart[1] = "Shoes"

print(cart)    # ['T-shirt', 'Shoes', 'Pen']
```

```
['T-shirt', 'Shoes', 'Pen']
```

Adding Items to a List

```
cart = ["T-shirt", "Lamp", "Pen"]

# Add "Book" to the list
cart.append("Book")

print(cart)    # ['T-shirt', 'Lamp', 'Pen', 'Book']
```

```
['T-shirt', 'Lamp', 'Pen', 'Book']
```

Add All the Items from Another List

```
cart = ["T-shirt", "Lamp", "Pen"]
fav_items = ["Headphones", "Phone"]

# Add all the items from fav_items to cart
cart.extend(fav_items)

print(cart)    # ['T-shirt', 'Lamp', 'Pen', 'Headphones', 'Phone']
```

```
['T-shirt', 'Lamp', 'Pen', 'Headphones', 'Phone']
```

insert an item at a specified position

```
cart = ["T-shirt", "Lamp", "Pen"]

# Add "Book" at index 2 (3rd position)
cart.insert(2, "Book")

print(cart)    # ['T-shirt', 'Lamp', 'Book', 'Pen']
```

```
['T-shirt', 'Lamp', 'Book', 'Pen']
```

Remove Items From a List

```
cart = ["T-shirt", "Lamp", "Pen", "Book"]

# Remove "Pen" from the list
cart.remove("Pen")    # ['T-shirt', 'Lamp', 'Book']

# Remove the last item
last_item = cart.pop()
print(cart)   # ['T-shirt', 'Lamp']
print(last_item)    # Book

# Clear the list
```

```
cart.clear()
print(cart)    # []
```

```
['T-shirt', 'Lamp']
Book
[]
```

delete individual items or even the whole list itself

```
cart = ['T-shirt', 'Lamp', 'Pen', 'Book']

# Delete the third item (index 2)
del cart[2]
print(cart)    # ['T-shirt', 'Lamp', 'Book']

# Delete the list itself
del cart
print(cart)    # NameError: name 'cart' is not defined
```

```
['T-shirt', 'Lamp', 'Book']
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_2211/548726998.py in <cell line: 0>()
      7 # Delete the list itself
      8 del cart
----> 9 print(cart)    # NameError: name 'cart' is not defined

NameError: name 'cart' is not defined
```

Next steps: Explain error

Copying a List

```
favorite_items = ["T-shirt", "Lamp", "Pen"]

cart = favorite_items

# Add an item to favorite_items list
favorite_items.append("Book")
```

```
print(f"favorite_items = {favorite_items}")
print(f"cart = {cart}")
```

```
favorite_items = ['T-shirt', 'Lamp', 'Pen', 'Book']
cart = ['T-shirt', 'Lamp', 'Pen', 'Book']
```

The len() Function

```
favorite_items = ["T-shirt", "Lamp", "Pen"]

size = len(favorite_items)
print(size)   # 3
```

```
3
```

List Membership Test

```
cart = ["T-shirt", "Lamp", "Pen"]

result = "Lamp" in cart
print(result)   # True

result = "Book" in cart
print(result)   # False
```

```
True
False
```

Iterating Through a List

```
cart_items = ["T-shirt", "Lamp", "Pen"]

for item in cart_items:
    print(item)
```

```
T-shirt
Lamp
Pen
```